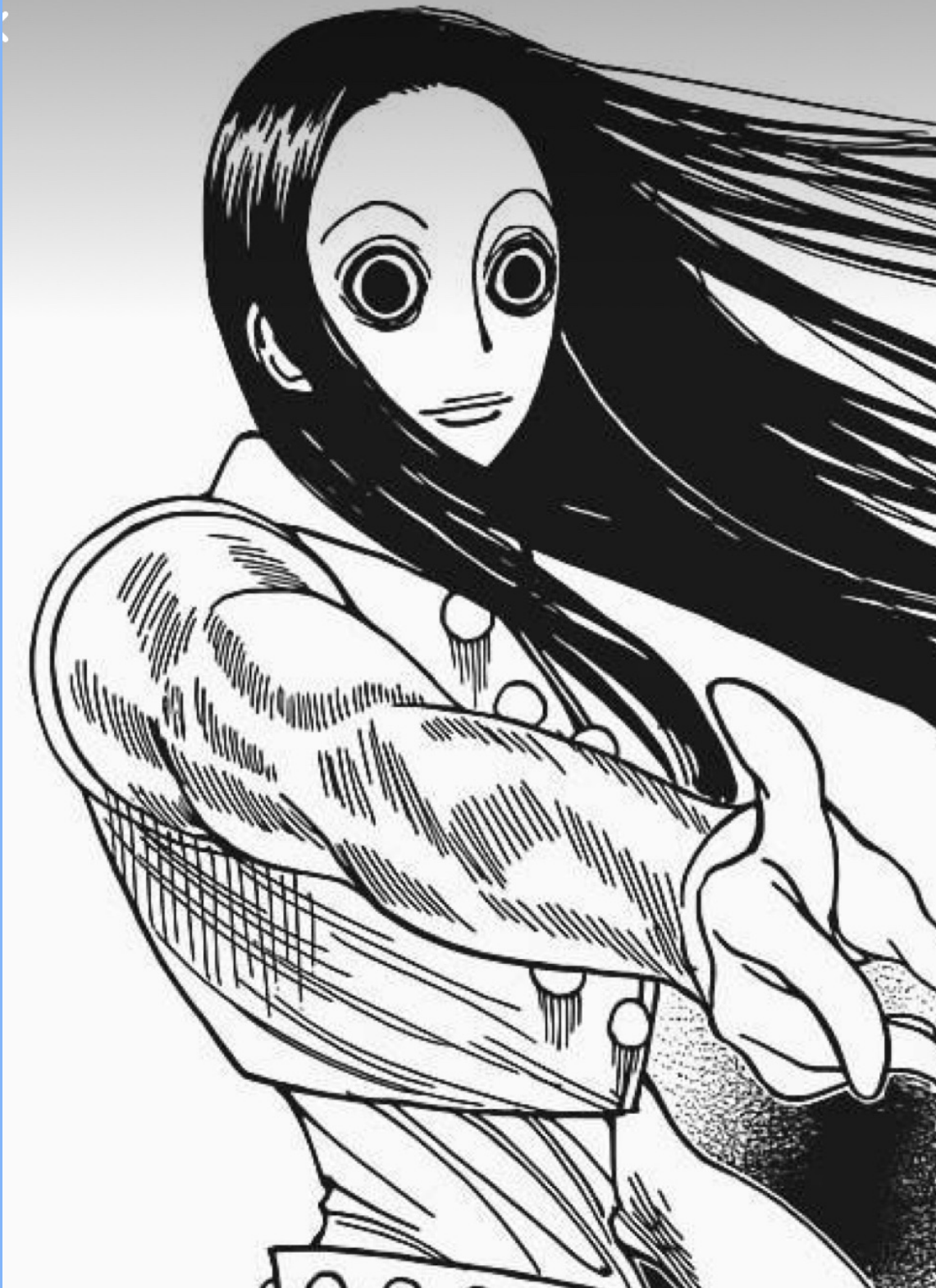

原点に強さ $p$ の電気双極子を固定したところ、$x$ 軸上の電界が $x>0$ で

$$E=\frac{p}{2\pi\varepsilon_0 x^3}$$ となった。

$t=0$ に $x=x_0\ (>0)$ の地点から電子（質量 $m$, 電荷 $-e$）を初速

$$v_0=\frac{1}{x_0}\sqrt{\frac{pe}{2\pi\varepsilon_0 m}}$$ で $x$ 軸方向に打ち出した。

時刻 $t$ における電子の位置 $x$ と速度 $v$ を求めよ。

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$$F=ma$$

$$m\frac{dv}{dt}=\vec{E}q$$

$$=-\frac{ep}{2\pi\varepsilon_0 x^3}$$

求めるものは $v$ と $x$

$ma=F$

$E=\frac{F}{q}$

$\overset{-e}{q}\cdot E=F$

$v=\frac{dx}{dt}$

$a=\frac{dv}{dt}$

$m\frac{dv}{dt}=-E\cdot e$

$=-\frac{pe}{2\pi\varepsilon_0 x^3}$

両辺に $v$ をかける

$$mv\frac{dv}{dt}=-\frac{evp}{2\pi\varepsilon_0 x^3}$$

$$=-\frac{ep}{2\pi\varepsilon_0 x^3}\cdot\frac{dx}{dt}$$

$$\int_0^t mv\frac{dv}{dt}dt=\int_0^x -\frac{ep}{2\pi\varepsilon_0 x^3}\frac{dx}{dt}dt$$

$$m\int_{v_0}^{v}\frac{d}{dt}\left(\frac{v^2}{2}\right)dt=-\frac{ep}{2\pi\varepsilon_0}\int_{x_0}^{x}\frac{1}{x^3}dx$$

$$\frac{mv^2}{2}-\frac{mv_0^2}{2}=\frac{ep}{2\pi\varepsilon_0}\left[\frac{1}{2x^2}\right]_{x_0}^{x}=\frac{ep}{4\pi\varepsilon_0}\left(\frac{1}{x^2}-\frac{1}{x_0^2}\right)$$

$1\cdot x^{1}\to\frac{1}{2}x^{2}$

$2\cdot x^{2}$

$x^{-3}$

$\frac{1}{2x^2}$

$\frac{1}{-2\cdot 1}=-\frac{1}{2}$

$$\frac{mv^2}{2} - \frac{m}{2}\left(\frac{1}{x_0^2}\cdot\frac{pe}{2\pi\varepsilon_0 m}\right) = \frac{ep}{4\pi\varepsilon_0}\left(\frac{1}{x^2}-\frac{1}{x_0^2}\right)$$

$$\frac{mv^2}{2} - \frac{\cancel{m}}{2}\left(\frac{1}{x_0^2}\cdot\frac{pe}{2\pi\varepsilon_0 \cancel{m}}\right) = \frac{1}{x^2}\cdot\frac{ep}{4\pi\varepsilon_0} - \frac{1}{x_0^2}\cdot\frac{ep}{4\pi\varepsilon_0}$$

$$\frac{mv^2}{2} - \underline{\left(\frac{1}{x_0^2}\cdot\frac{pe}{4\pi\varepsilon_0}\right)} = \left(\frac{1}{x^2}\cdot\frac{ep}{4\pi\varepsilon_0}\right) - \underline{\left(\frac{1}{x_0^2}\cdot\frac{ep}{4\pi\varepsilon_0}\right)}$$

$$\frac{mv^2}{2} = \frac{1}{x^2}\cdot\frac{ep}{4\pi\varepsilon_0}$$

$$v^2 = \frac{\cancel{2}ep}{\cancel{4}_{2}\pi\varepsilon_0 m x^2}$$

$$\underline{v = \sqrt{\frac{ep}{2\pi\varepsilon_0 m}} \times \frac{1}{x}}$$

$$\frac{dx}{dt} = \frac{1}{x}\sqrt{\frac{ep}{2\pi\varepsilon_0 m}}$$

$$x\frac{dx}{dt} = \sqrt{\frac{ep}{2\pi\varepsilon_0 m}}$$

$$\int_0^t x\frac{dx}{dt}\,dt = \int_0^t \sqrt{\frac{ep}{2\pi\varepsilon_0 m}}\,dt$$

$$\int_{x_0}^{x} x\,dx = \int_0^t \sqrt{\frac{ep}{2\pi\varepsilon_0 m}}\,dt$$

$$\left[\frac{1}{2}x^2\right]_{x_0}^{x} = \left[\sqrt{\frac{ep}{2\pi\varepsilon_0 m}}\,t\right]_0^t$$

$$\frac{1}{2}x^2 - \frac{1}{2}x_0^2 = \sqrt{\frac{ep}{2\pi\varepsilon_0 m}}\,t$$

$$x^2 - x_0^2 = 2t\sqrt{\frac{ep}{2\pi\varepsilon_0 m}}$$

$$x^2 = x_0^2 + 2t\sqrt{\frac{ep}{2\pi\varepsilon_0 m}}$$

$$\underline{x = \sqrt{x_0^2 + 2t\sqrt{\frac{ep}{2\pi\varepsilon_0 m}}}}$$

(1)

$$F = ma$$

$$E = \frac{F}{q} \Leftrightarrow F = Eq$$

$$\therefore ma = Eq$$

$$m\frac{dv}{dt} = \frac{ep}{2\pi\varepsilon_0 x^3} \quad \times v$$

$$mv\frac{dv}{dt} = \frac{ep}{2\pi\varepsilon_0 x^3} \cdot \frac{dx}{dt}$$

$$m\int_0^t v\frac{dv}{dt}dt = \int_0^t \frac{ep}{2\pi\varepsilon_0 x^3}\frac{dx}{dt}dt$$

$$m\int_{v_0}^{v} v\,dv = \frac{ep}{2\pi\varepsilon_0}\int_{x_0}^{x}\frac{1}{x^3}dx$$

$$m\left[\frac{v^2}{2}\right]_{v_0}^{v} = \frac{ep}{2\pi\varepsilon_0}\left[\frac{1}{2x^2}\right]_{x_0}^{x}$$

$$\frac{mv^2}{2} - \frac{mv_0^2}{2} = \frac{pe}{2\pi\varepsilon_0}\left(\frac{1}{2x^2} - \frac{1}{2x_0^2}\right)$$

$$\frac{mv^2}{2} - \underbrace{\frac{m}{2}\left(\frac{1}{x_0^2} \times \frac{pe}{2\pi\varepsilon_0 m}\right)}_{\times} = \underbrace{\frac{pe}{2\pi\varepsilon_0}}_{\lambda}\left(\frac{1}{2x^2} - \underbrace{\frac{1}{2x_0^2}}_{\infty}\right)$$

$$\frac{mv^2}{2} = \frac{1}{x^2} \times \frac{pe}{2\pi\varepsilon_0}$$

$$\frac{mv^2}{2} = \frac{1}{x_2} \times \frac{pe}{2\pi\varepsilon_0}$$

$$v^2 = \frac{\cancel{2}pe}{{}_2\cancel{4}\pi\varepsilon_0 x_2 m} = \frac{pe}{2\pi\varepsilon_0 m} \times \frac{1}{x_2}$$

$$\therefore v = \sqrt{\frac{pe}{2\pi\varepsilon_0 m}} \times \frac{1}{x}$$

$$\frac{dx}{dt} = \sqrt{\frac{pe}{2\pi\varepsilon_0 m}} \times \frac{1}{x}$$

$$x\frac{dx}{dt} = \sqrt{\frac{pe}{2\pi\varepsilon_0 m}}$$

$$\int_0^t x\frac{dx}{dt}\,dt = \int_0^t \sqrt{\frac{pe}{2\pi\varepsilon_0 m}}\,dt$$

$$\int_{x_0}^{x} x\,dx = \int_0^t \sqrt{\frac{pe}{2\pi\varepsilon_0 m}}\,dt$$

$$\left[\frac{x^2}{2}\right]_{x_0}^{x} = \left[\sqrt{\frac{pe}{2\pi\varepsilon_0 m}}\,t\right]_0^t$$

$$\frac{1}{2}(x^2 - x_0^2) = \sqrt{\frac{pe}{2\pi\varepsilon_0 m}}\,t$$

$$x^2 - x_0^2 = 2t\sqrt{\frac{pe}{2\pi\varepsilon_0 m}}$$

$$x^2 = x_0^2 + 2t\sqrt{\frac{pe}{2\pi\varepsilon_0 m}}$$

$$\therefore x = \sqrt{x_0^2 + 2t\sqrt{\frac{pe}{2\pi\varepsilon_0 m}}}$$

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$$z^2 = \frac{Pe}{2\pi\varepsilon_0 m} \times \frac{1}{x_0^2 + 2t\sqrt{\frac{Pe}{2\pi\varepsilon_0 m}}}$$

$$= \frac{Pe}{2\pi\varepsilon_0 m x_0^2 + 4\pi\varepsilon_0 m t\sqrt{\frac{Pe}{2\pi\varepsilon_0 m}}}$$

$$= \frac{eP}{4ept^2 - 2\pi m x_0^4 \varepsilon_0}\left(2\sqrt{\frac{eP}{2\pi\varepsilon_0 m}}\,t - x_0^2\right)$$

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